COMICCRAFT PROJECT DOCUMENTATION

Phase 5: Comic Generation Workflow

Panel Creation, Image Synthesis, Dialogue Overlay & Interactive Preview

1. Phase Overview

The fifth phase of the ComicCraft project focuses on implementing the complete comic generation workflow. In this phase, the AI-generated story, characters, scenes, dialogues, and generated images are synthesized to construct complete, publication-ready comic panels.

2. Objectives

The primary objectives of Phase 5 are to:

- Convert AI-generated story beats into individual, structured comic panels.
- Trigger image synthesis pipelines for each specific scene description.
- Integrate characters, contextual dialogues, and narrative captions into panels.
- Maintain strict visual character consistency across multi-panel layouts.
- Arrange panels sequentially in responsive layout grids.
- Provide a dynamic comic preview and individual panel regeneration system.

3. Complete Comic Generation Workflow

The end-to-end operational pipeline progresses through the following stages:

```
User Prompt → AI Story Generation → Character Generation → Scene Generation → Dialogue Generation → Panel Creation → Image Generation → Dialogue & Caption Placement → Comic Assembly → Comic Preview
```

4. Story to Comic Conversion & Panel Architecture

The narrative is segmented into discrete scenes mapped directly to visual panels:

```
Story → Scene 1 (Panel 1) | Scene 2 (Panel 2) | Scene 3 (Panel 3) | Scene 4 (Panel 4)
```

Structure of an Individual Panel Module:

```
Panel Structure
├── Scene Metadata
├── Character Entities & Positions
├── Action Vectors
├── Background Environment
├── Speech / Thought Dialogues
└── Narrative Captions
```

5. Image Generation & Character Consistency

Every scene description triggers an optimized prompt sent to the Image Generation API:

```
Scene Description → Image Prompt → Image Generation API → Rendered Image → Comic Panel
```

Maintaining Character Consistency Across Panels:

To ensure uniform character identity, standard visual seeds and consistent descriptors are injected across all panel prompts:

- **Appearance:** Facial structures, hair style, hair color.
- **Attire:** Fixed color palettes and clothing style.
- **Art Style:** Enforced rendering parameters (e.g., cartoon, anime, vintage comic book).

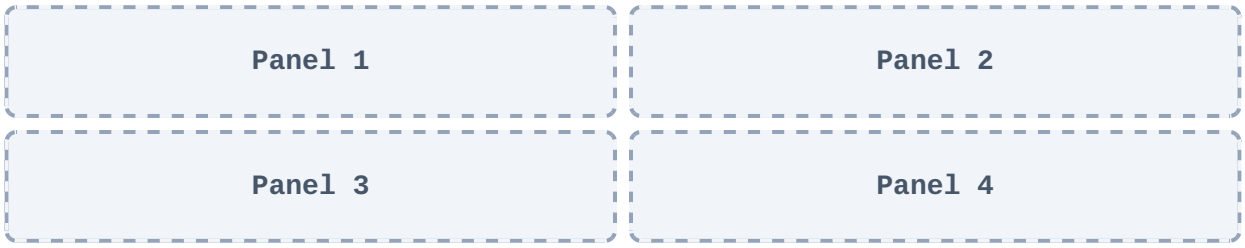
6. Dialogue & Caption Overlay System

The rendering layer overlays typography components on generated background images:

- **Speech Bubbles:** Pointed towards active speaker characters.
- **Thought Bubbles:** Used for internal monologues.
- **Narration Boxes & Captions:** Positioned at top/bottom margins for story transitions.

7. Panel Grid Layout & Arrangement

Panels are organized dynamically according to user selection (e.g., 2x2 grid for 4-panel stories):



8. Interactive Preview & Panel Regeneration

Users can review their generated comic prior to final export. If a panel does not meet expectations, the single panel can be re-rendered without regenerating the full comic:

```
Select Panel → Click Regenerate → Generate New Image/Content → Live Panel Swap
```

9. Error Handling & Fault Tolerance

The workflow handles pipeline bottlenecks gracefully:

- Image API failures trigger automatic retry loops or fallback placeholder graphics.
- Missing dialogues or malformed AI outputs are caught by schema validators.
- Timeouts provide clear user feedback with option to resume or retry.

10. Phase 5 Deliverables & Status

- Story-to-Panel Parsing Algorithm
- Scene Image Generation Engine
- Character Consistency Enforcement Pipeline
- Speech Bubble & Caption Overlay Component
- Dynamic Grid Arrangement System
- Full Comic Interactive Preview Workspace
- Single Panel Regeneration Subsystem
- Comprehensive Fault Handling Middleware

Current Phase Status: COMPLETED / READY FOR PHASE 6